

MANIPAL UNIVERSITY JAIPUR
Department of Computer Science & Engineering
Course Hand-out

A. Basic Details:

Programme Name:	B.Tech. CSE
Course Name:	Data Structures and Algorithms
Course Code:	CSC2101
LTPC (<i>Lecture Tutorial Practical Credits</i>):	3 1 0 4
Session:	July-November 2026
Class:	B.Tech. CSE 2nd Year III Semester (Core Course)
Course Coordinator:	Dr. Ganpat Singh Chauhan Cluster Coordinators: Dr. Neha Janu, Dr. Pramod Singh Rathore, Dr Pankaj Updhyay, Dr. Manoj Bohra, Dr. Rekha Chaturvedi
Course Instructor(s):	Dr. Amit Kumar Gupta, Dr. Mahesh Jangid, Dr. Shikha Mundra, Dr. Amit Garg, Dr. Satpal Singh Kushwaha, Dr. Sayar Singh Shekhawat, Dr. Tarun Jain, Dr. Amit Kumar Sharma, Mr. Aditya Narayan Hati, Dr. Reddy Mounika Bommisetty, Mr. Virendra Dehru, Dr. Rahul Saxena, Dr. Shweta Sharma, Dr. Avani Sharma, Dr. Pankaj Vyas, Dr. Manish Rai, Dr. Nikhil Pateria, Dr. Siddharth Kumar, Dr. Neha Mahala, Mr. Ravindra Kumar Soni, Dr. Shivendra Dubey, Dr. Prashant Hemrajani, Dr. Govind Chhimpa, Dr. Rohit Mittal, Dr. Bhawana Sharma, Ms. Soni Gupta, Dr. Malvinder Singh Bali, Dr. Anil Kumar Prajapati, Dr. Gauransh Kalla, Ms. Meenakshi Gaur
Additional Practitioner(s) – if any (<i>Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.</i>):	Industry Expert Talk, Programming Competition

B. Introduction: Data Structures and Algorithms (DSA) is a fundamental course that provides the knowledge and skills required to design efficient solutions for computational problems. The course introduces the concepts of abstract data types, linear and non-linear data structures, and algorithm design techniques. Students learn to implement and analyse various data structures such as trees, graphs, heaps, and hashing, along with algorithmic paradigms including recursion and divide-and-conquer. Emphasis is placed on evaluating algorithm efficiency using time and space complexity analysis, asymptotic notations, and recurrence relations. Through theoretical concepts, programming demonstrations, and problem-solving exercises, the course develops students' analytical thinking and programming skills essential for software development and advanced computing applications.

C. Course Outcomes: At the end of the course, students will be able to:

<i>CO</i>	<i>CO Statement</i>	<i>Level</i>	<i>Target Attainment %</i>	<i>Target Attainment level</i>
[CSC2101.1]	Compare Iterative and recursive algorithms and describe the criteria for a well-defined algorithm	Analyse (L4)	80%	3
[CSC2101.2]	Analyze tree and heap data structures, including binary trees, AVL trees, B-trees, red-black trees, and priority queues	Analyse (L4)	70%	2
[CSC2101.3]	Apply graph representations and traversal algorithms such as BFS and DFA to solve real-world problems	Apply (L3)	80%	3
[CSC2101.4]	Analyse and evaluate algorithms using divide and conquer and greedy strategies for problems like sorting, matrix multiplication, and shortest path computation	Analyse (L4)	70%	2
[CSC2101.5]	Compare and select appropriate algorithm techniques and data structures to optimize the performance of computational tasks	Analyse (L4)	70%	2

- Bloom's Level: Cognitive level of Revised Bloom's Taxonomy
- Target Attainment Level: Tentative/ Estimated target attainment for each CO. Information about Target Attainment Levels is mentioned as follows:

Attainment (%)	Level
< 60 %	0
≥ 60% < 70%	1
≥ 70% < 80%	2
≥ 80	3

D. Program Outcomes and Program Specific Outcomes:

[PO.1].	Engineering knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
[PO.2].	Problem analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
[PO.3].	Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
[PO.4].	Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
[PO.5].	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).
[PO.6].	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
[PO.7].	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
[PO.8].	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
[PO.9].	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
[PO.10].	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to One's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
[PO.11].	Life-Long Learning: Recognize the need for and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).
[PSO.1].	Design and develop optimized computational solutions using appropriate programming paradigms, contributing to sustainable development to ensure efficient resource utilization
[PSO.2].	Create intelligent or innovative solutions to real-world challenges by applying modern tools, software engineering practices, and emerging technologies
[PSO.3].	Process and communicate data effectively while ensuring software security, data privacy, and reliable data communication in practical applications.

E. Assessment Plan: Clearly write the criteria, its description, and associated marks for assessment of student achievements. In addition, the attendance requirements, assignments need to be mentioned in this section.

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Class Work Sessional (CWS): 4 Quizzes (Best of 3): 15 marks 2 Assignment: 5 + 5 = 10 marks (Assignment Evaluation scheme will be shared separately) - Attendance: 5 marks ($\geq 95\%$: 5, $90\% - < 95\%$: 4, $85\% - < 90\%$: 3, $80\% - < 85\%$: 2, $75\% - < 80\%$: 1, $< 75\%$: 0) OR If any student clear NPTEL certification then 25 marks can be awarded directly in place of Assignments/Quizzes marks: - NPTEL certification: 25 marks NPTEL Course (Registration deadline: 27-07-26) 1. Data Structure and Algorithms using Java NPTEL Link: https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs162 - Attendance: 5 marks ($\geq 95\%$: 5, $90\% - < 95\%$: 4, $85\% - < 90\%$: 3, $80\% - < 85\%$: 2, $75\% - < 80\%$: 1, $< 75\%$: 0) Note: if no NPTEL certification then marks will be awarded based on quizzes, assignments, and attendance/participation)	30 (15 + 10 + 5)
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

F. Syllabus: Introduction: Algorithm definition and criteria of algorithms, iterative and recursive algorithms; **Tree:** Types of trees, representation of binary tree using array and linked list, binary search tree, different operations (recursive and non-recursive), red-black trees; Augmenting Data Structures, AVL trees, B-tree; **Heap:** Heap Sort; **Graph:** Introduction, representation, operations and applications; **Graph Search Algorithm:** Breadth First Search and Depth First Search; **Indexing and Hashing. Performance Analysis:** Priori and Posteriori Analysis, asymptotic notations, space complexity, time complexity, performance measurement of iterative and recursive algorithms; **Solving recurrence relations:** Substitution method, iterative method, recursive tree method, Master Theorem; Analysis of Different algorithms studied in earlier courses like Sorting, Searching, and algorithms studied with strategies like Divide and Conquer Algorithms, and Greedy algorithms; **Divide and Conquer:** Introduction, Binary Search, Finding Maximum and Minimum, Merge Sort, Quick Sort, large integer multiplication, Strassen's matrix multiplication.

Textbook(s) and/or Reference Book(s):

1. A. M. Tenenbaum, Y. Langsam, and M. J. Augenstein, Data Structures Using C, (1e), New Delhi, India: Pearson Education, 2019.
2. E. Horowitz, S. Sahni, and S. Anderson-Freed, Fundamentals of Data Structures in C, (2e), Hyderabad, India: Universities Press, 2022.
3. B. A. Forouzan and R. F. Gilberg, Computer Science: A Structured Programming Approach Using C, (3e), Boston, MA: Cengage Learning, 2007.
4. TH Cormen, CF Leiserson, RL Rivest, C Stein, Introduction to Algorithms, 3rd Ed., MIT Press, 2009.

G. Lecture Plan:

Class No.	Topics	Session Outcome	Corresponding CO	Mode of Delivery	Mode of Assessing the Outcome
1	Course Handout Discussion	Explain the course objectives, syllabus, assessment scheme, course outcomes, and expected learning outcomes	CSC2101.1	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
2	Introduction to Data Structures, Abstract Data Types (ADTs), Linear and Non-Linear Data Structures	Explain the concepts of data structures and abstract data types, classify data structures, and identify their application	CSC2101.1	Group-teaching and learning	MTE, Quiz-1, Assignment-1, ETE
3	Recap of Arrays and Linked Lists	Explain operations on arrays and linked lists and compare their characteristics	CSC2101.1	Group-teaching and learning	MTE, Quiz-1, Assignment-1, ETE
4	Recap of Stack and Queue	Explain stack and queue operations and identify appropriate applications of each data structure	CSC2101.1	Group-teaching and learning	MTE, Quiz-1, Assignment-1, ETE
5	Introduction to Algorithms: Definition, Characteristics, Iterative and Recursive Algorithms and Algorithm Design Criteria	Explain the characteristics of algorithms, distinguish iterative and recursive approaches, and identify the criteria for efficient algorithm design	CSC2101.1	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
6	Algorithm Design using Iterative and Recursive Approaches (Tower of Hanoi)	Develop algorithms using iterative and recursive approaches and compare their implementations through suitable examples	CSC2101.1	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
7	Introduction to Trees, Tree Terminology, Types of Trees	Explain tree concepts, terminology, applications and differentiate various tree structures	CSC2101.2	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
8	Binary Tree Representation using Arrays	Implement binary tree representation using arrays and perform basic operations	CSC2101.2	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
9	Binary Tree Representation using Linked Lists	Implement binary tree representation using linked structures and perform basic operations	CSC2101.2	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
10	Binary Search Tree (BST): Properties, Insertion in BST (Recursive and Non-Recursive)	Explain the properties of Binary Search Trees and implement insertion operations using recursive and non-recursive approaches	CSC2101.2	Learning through problem-solving	MTE, Quiz-1, Assignment-1, ETE

11	BST Operations: Deletion (Recursive and Non-Recursive)	Implement deletion operations in Binary Search Trees using recursive and non-recursive approaches	CSC2101.2	Learning through problem-solving	MTE, Quiz-1, Assignment-1, ETE
12	BST Operations: Searching (Recursive and Non-Recursive)	Implement recursive and non-recursive searching techniques	CSC2101.2	Learning through problem-solving	MTE, Quiz-1, Assignment-1, ETE
13	BST Operations: Traversal (Recursive and Non-Recursive)	Implement recursive and non-recursive tree traversal techniques	CSC2101.2	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
14	AVL Trees: Introduction and Rotations	Explain AVL tree properties and perform tree balancing using rotations	CSC2101.2	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
15	AVL Trees: Insertion and Deletion	Implement insertion and deletion operations in AVL Trees	CSC2101.2	Learning through problem-solving	MTE, Quiz-1, Assignment-1, ETE
16	Augmenting Data Structures: Red-Black Trees Introduction	Explain the properties and balancing mechanism of Red-Black Trees	CSC2101.2	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
17	Red-Black Trees: Insertion and Deletion	Implement insertion and deletion operations in Red-Black Trees	CSC2101.2	Learning through problem-solving	MTE, Quiz-1, Assignment-1, ETE
18	B-Trees: Properties and Operations	Explain the properties of B-Trees and perform basic operations	CSC2101.2	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
19	B-Trees: Insertion and Deletion	Implement insertion and deletion operations in B-Trees	CSC2101.2	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
20	Comparative Study and Applications of Trees	Compare different tree data structures and identify suitable applications	CSC2101.2	Flipped Classroom	MTE, Quiz-1, Assignment-1, ETE
21	Heap: Introduction, Max/Min Heap, Build Heap, Heap Operations	Explain heap properties, construct Min/Max Heaps, and perform heap operations	CSC2101.2	Lecture based Teaching Learning	MTE, Quiz-1, Assignment-1, ETE
22	Heap Sort; Application and analysis	Implement Heap Sort and analyse its computational complexity	CSC2101.2	Learning through problem-solving	MTE, Quiz-1, Assignment-1, ETE
23	Introduction to Graphs and basic types of graphs	Explain graph terminology, classify graph types, and identify graph applications	CSC2101.3	Lecture based Teaching Learning	MTE, Quiz-2, Assignment-1, ETE
24	Graph Representation and Operations	Implement graph representations using adjacency matrix and adjacency list	CSC2101.3	Lecture based Teaching Learning	MTE, Quiz-2, Assignment-1, ETE
25	Applications of Graphs	Apply graph concepts to model and solve practical computational problems	CSC2101.3	Group-teaching and learning	MTE, Quiz-2, Assignment-1, ETE
26	Breadth First Search (BFS): finding shortest path	Implement Breadth First Search (BFS) traversal and analyse its applications	CSC2101.3	Learning through problem-solving	MTE, Quiz-2, Assignment-1, ETE
27	Depth First Search (DFS): find path exists between two specified vertices	Implement Depth First Search (DFS) traversal and analyse its applications	CSC2101.3	Learning through problem-solving	MTE, Quiz-2, Assignment-1, ETE

28	Comparison and Applications of BFS and DFS with Examples	Compare Breadth First Search and Depth First Search based on their characteristics and applications	CSC2101.3	Group-teaching and learning	MTE, Quiz-2, Assignment-1, ETE
29	Indexing Techniques	Explain indexing techniques and their role in efficient data retrieval	CSC2101.3	Flipped Classroom	MTE, Quiz-2, Assignment-1, ETE
30	Hashing and Hash Functions	Explain hashing concepts and implement suitable hash functions	CSC2101.3	Lecture based Teaching Learning	MTE, Quiz-2, Assignment-1, ETE
31	Collision Resolution Techniques: linear probing, quadratic probing, double hashing, and separate chaining	Apply collision resolution techniques such as linear probing, quadratic probing, double hashing, and separate chaining to efficiently handle hash collisions	CSC2101.3	Lecture based Teaching Learning	MTE, Quiz-2, Assignment-1, ETE
32	Revision and Mid Term Examination Discussion	Comprehensive Revision of MTE Course	CSC2101.3	Individual learning/ self-study	MTE, Quiz-2, Assignment-1, ETE
MID-TERM EXAMINATION					
33	Introduction to Performance Analysis: Priori and Posteriori Analysis	Explain the need for algorithm performance analysis and distinguish between a priori and a posteriori analysis	CSC2101.5	Lecture based Teaching Learning	Quiz-3, Assignment-2, ETE
34	Time and Space Complexity	Analyze algorithms using time and space complexity measures	CSC2101.5	Lecture based Teaching Learning	Quiz-3, Assignment-2, ETE
35	Asymptotic Notations (Big-O, Big-Ω, Big-Θ)	Apply asymptotic notations to evaluate algorithm efficiency	CSC2101.5	Lecture based Teaching Learning	Quiz-3, Assignment-2, ETE
36	Performance Analysis of Iterative Algorithms	Analyze iterative algorithms using appropriate complexity measures	CSC2101.5	Lecture based Teaching Learning	Quiz-3, Assignment-2, ETE
37	Performance Analysis of Recursive Algorithms	Formulate and analyze recurrence relations for recursive algorithms	CSC2101.5	Lecture based Teaching Learning	Quiz-3, Assignment-2, ETE
38	Solving Recurrence Relations: Substitution Method	Solve recurrence relations using the substitution method	CSC2101.5	Learning through problem-solving	Quiz-3, Assignment-2, ETE
39	Iteration Method	Solve recurrence relations using the iteration method	CSC2101.5	Learning through problem-solving	Quiz-3, Assignment-2, ETE
40	Recursion Tree Method	Solve recurrence relations using the recursion tree method	CSC2101.5	Learning through problem-solving	Quiz-3, Assignment-2, ETE
41	Master Theorem	Apply the Master Theorem to analyze divide-and-conquer algorithms	CSC2101.5	Learning through problem-solving	Quiz-3, Assignment-2, ETE

42	Introduction to Divide and Conquer and Greedy Approach	Explain the Divide-and-Conquer and Greedy algorithmic paradigms, distinguish their characteristics, and identify suitable application domains	CSC2101.4	Individual learning/ self-study	Quiz-4, Assignment-2, ETE
43	Divide and Conquer: Introduction, Binary Search, Finding Maximum & Minimum	Apply the divide-and-conquer paradigm to implement and analyze Binary Search and Maximum-Minimum algorithms	CSC2101.4	Lecture based Teaching Learning	Quiz-4, Assignment-2, ETE
44	Divide and Conquer: Merge Sort	Implement Merge Sort using the divide-and-conquer paradigm and analyze its performance	CSC2101.4	Group-teaching and learning	Quiz-4, Assignment-2, ETE
45	Divide and Conquer: Quick Sort	Implement Quick Sort using the divide-and-conquer paradigm and analyze its performance	CSC2101.4	Group-teaching and learning	Quiz-4, Assignment-2, ETE
46	Large Integer Multiplication	Implement the divide-and-conquer algorithm for large integer multiplication and analyze its efficiency	CSC2101.4	Flipped Classroom	Quiz-4, Assignment-2, ETE
47	Strassen's Matrix Multiplication	Implement Strassen's Matrix Multiplication algorithm and analyze its computational complexity	CSC2101.4	Flipped Classroom	Quiz-4, Assignment-2, ETE
48	Revision and Problem Solving	Integrate course concepts through comprehensive revision and problem-solving in preparation for the End-Term Examination	CSC2101.4	Individual learning/ self-study	Quiz-4, Assignment-2, ETE
END-TERM EXAMINATION					

Outside class engagement activities (To complete the 30 Notional hours for each credit as per NCrF guidelines):

Type of Activity	Difficulty Level	Hours Engaged
Watching recommended video lectures and tutorials to recap data structures concepts	Easy	4
Extra Class on Array	Easy	2
Extra Class on Linked List	Easy	2
Extra Class on Stack	Easy	2

Extra Class on Queue	Easy	2
Participating in academic clubs, technical competitions, or seminars	Medium	10
Participating in faculty-guided doubt-clearing and concept-reinforcement sessions	Easy	6
Mini research projects, project documentation, or presentations	High	6
One-on-one remedial sessions, tutorials, or simplified handouts	Low	8
Visiting laboratories or project labs for extended hands-on programming practice	Medium	8
Exploring relevant online courses, MOOCs, or certification programs (e.g., NPTEL)	Medium	10
Participation in hackathons or innovation challenges (departmental or inter-college level)	High	6
Algorithm Design competition	Medium	6

H. Course Articulation Matrix:

CO	STATEMENT	Correlation with Program Outcomes and Program Specific Outcomes													
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
[CSC2101.1]	Compare Iterative and recursive algorithms and describe the criteria for a well-defined algorithm	3	3	2	3	3	2	0	0	0	0	0	2	3	3
[CSC2101.2]	Analyze tree and heap data structures, including binary trees, AVL trees, B-trees, red-black trees, and priority queues	3	3	2	1	3	2	0	0	0	0	0	2	3	2
[CSC2101.3]	Apply graph representations and traversal algorithms such as BFS and DFA to solve real-world problems	3	3	3	3	3	2	0	0	0	0	0	2	3	3
[CSC2101.4]	Analyse and evaluate algorithms using divide and conquer and greedy strategies for problems like sorting, matrix multiplication, and shortest path computation	3	3	1	2	3	3	0	0	0	0	0	2	3	3
[CSC2101.5]	Compare and select appropriate algorithm techniques and data structures to optimize the performance of computational tasks	3	3	3	2	3	3	0	0	0	0	0	3	3	3